

SURBHIT PRATIK

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EDUCATION

University of Southern California

Master of Science (MS), Computer Science

January 2024 - December 2025

SRM Institute of Science and Technology

Bachelor of Technology (B.Tech), Computer Science; Minor: Software Engineering

July 2019 - June 2023

PROFESSIONAL EXPERIENCE

Easley Dunn Productions, Inc

Software Engineer

February 2026 - Present

Santa Clara, CA, USA

- Engineered an **LLM-driven analytics pipeline** for automated reporting and near real-time telemetry over **GCP BigQuery** and **Firebase**, with schema-grounded execution **guardrails** and a **conversational agent** interface, automating ~80% of ad-hoc requests
- Architected a stateful memory retention system using dynamic context compression to maintain multi-turn context; **reduced API token consumption** by ~35% and lowered inference costs for complex analytical follow-ups
- Shipped an **MCP server** on **Google Cloud Run** exposing 3 custom tools that index fragmented internal data (Firebase, GitHub, meeting transcripts) behind a single retrieval interface, eliminating context-switching during documentation and repo exploration
- Implemented **Firebase Analytics** event tracking (C#) across 70+ gameplay events in a **Unity** application, routing 12 user flows into **Looker Studio** dashboards to visualize 17 key metrics for the product and design teams

Amphenol

Software Engineering Intern

May 2025 - August 2025

St. Marys, PA, USA

- Architected a lightweight Internal Developer Portal using a **Python webhook engine**, integrating the GitHub and Jira **REST APIs** to automate repository provisioning and enforce naming governance, cutting project setup time by >30%
- Built a **RESTful Python microservice** to automate hardware component nomenclature, replacing unstructured legacy workflows with strict naming schemas that reduced downstream quoting errors by 23%
- Built a C# WinForms dashboard to visualize test results from **SQL Server**, speeding root-cause analysis and eliminating significant manual diagnostic effort

HighRadius

Software Developer Intern

January 2022 - April 2022

Remote

- Engineered a scalable loan prediction platform using a **React** and **Node.js** stack, cutting **MySQL** data retrieval latency by 25% through optimized database indexing and server-side pagination
- Integrated **Random Forest** and **Gradient Boosting** algorithms to develop predictive ML models, achieving a 14.3% reduction in loan repayment discrepancies and improving overall financial **system reliability**

Coal India Ltd

Software Developer Intern

September 2021 - October 2021

Ranchi, India

- Developed and deployed a user-friendly accommodation search application leveraging **JavaScript** while utilizing **Terraform** for infrastructure as code to streamline deployment processes
- Designed and implemented a backend with **Django** and **PostgreSQL**, containerized it with **Docker**, and integrated **AWS S3** and **RDS**, which improved data reliability and security

PROJECTS

Mia: Real-Time Voice Agent for Google Meet [Demo](#)

- Built an LLM-based voice agent that joins Google Meet, pairing **streaming STT** with **Claude Opus** over **rolling call state** to perform in-call queries through concurrent Calendar and Meet tool calls and emit structured summaries ~**2.9s** p95 speech-to-speech
- Enforced LLM **guardrails** through a lightweight **intent classifier** conditioned on rolling meeting state, run asynchronously to suppress **prompt injection** and out-of-scope completions at <**150ms** added p95

CodeSense: Agentic Codebase Retrieval and Chat [Demo](#)

- Developed an LLM-based coding agent (**Claude Sonnet 5**) paired with a custom **AST-based RAG system** for precise context retrieval, parsing and mapping complex dependency chains across codebases spanning 10,000+ interdependent files
- Implemented multiple retrieval methodologies, **SQLite FTS BM25** and **Abstract Syntax Trees**, for efficient and accurate code indexing and lazy lookups, saving model context by **74%**

Made AI play Mafia: Multi-agent asynchronous communication [Code](#)

- Developed **async multi-agent** AI system, enabling structured communication among 6+ autonomous agents in social deduction gameplay scenarios
- Implemented modular 2-part brain architecture (Scheduler & Generator), with a **concurrency-safe** shared context

SKILLS

Languages: Python, C/C++, C#, Java, JavaScript, SQL

AI/LLM: Claude Agent SDK, OpenAI Agents SDK, Tool Calling, MCP Server, LangGraph, Deepgram, ElevenLabs, Pydantic, HuggingFace

Backend & Web: REST APIs, Django, Flask, React, Asynchronous Programming (asyncio)

Cloud & Data: GCP, AWS, Docker, Kubernetes, Terraform, Git, MySQL, PostgreSQL, Jira, BigQuery, Firebase, S3, EC2, RDS